

Methodology: Pipeline Design and Formal Definitions

This section describes the five-stage computational meta-analysis pipeline. Each stage corresponds to a tested, independently executable script that reads upstream outputs and produces structured artifacts. The pipeline extends the systematic literature analysis approach of Knight et al. knight2022sep—which combined manual annotation with ontology-based automated analysis—by substituting manual coding with fully automated, LLM-driven assertion extraction and citation-weighted hypothesis scoring. All code, configuration files, and reproducibility instructions—including a Dockerized execution environment to guarantee dependency isolation—are publicly available in the project repository (https://github.com/ActiveInferenceInstitute/literature_met)

Pipeline Overview

The five-stage pipeline is summarized in `tab:pipeline_stages`.

Table 1: Five-stage computational meta-analysis pipeline. Each stage corresponds to an independently executable script that reads upstream outputs and produces structured artifacts. Cross-references link to detailed methodology sections